

ProjectBoard

A self-hosted, kanban-style project management system

This document provides an overview of ProjectBoard's features and functions, covering the core kanban workflow, collaboration tools, reporting, and AI-assisted features built into the system. Each section is illustrated with real screenshots of the system.

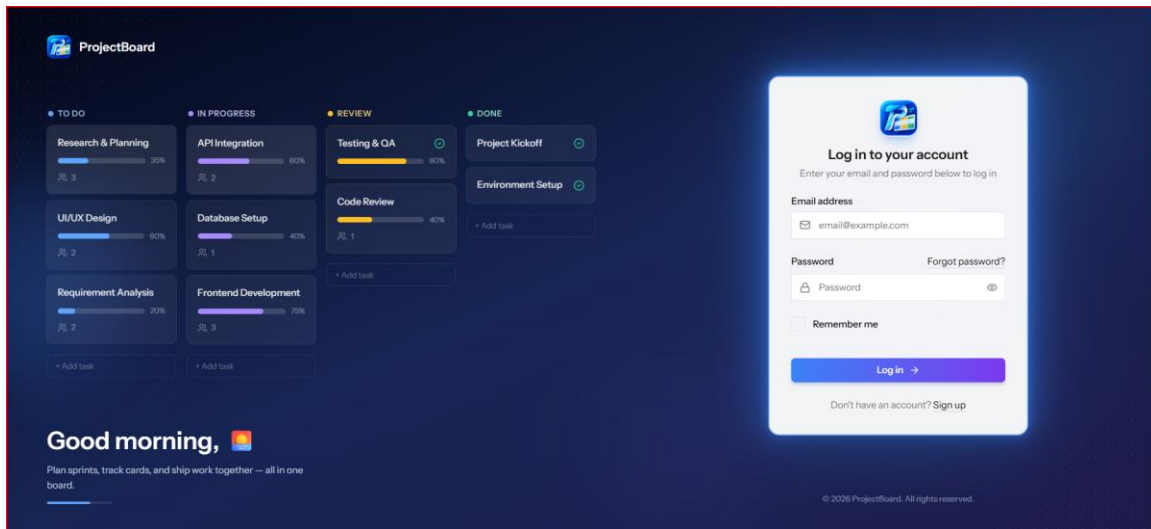


Figure 1: ProjectBoard login page

1. Introduction

ProjectBoard is a Trello-style kanban project management application built with Laravel and Vue.js. It allows teams to organise work into workspaces, boards, lists, and cards, track progress through checklists, collaborate through comments and chat, and generate detailed reports — without the recurring subscription costs of commercial alternatives.

Core structure:

- Workspace — a container for a team or department's boards.
- Board — a kanban board belonging to a workspace.
- List — a column on a board (e.g. To Do, In Progress, Done).
- Card — an individual task within a list.
- Checklist — a set of sub-tasks (items) within a card.

2. Workspaces and Boards

Workspaces group related boards together. Each workspace and board tile shows a live checklist-completion progress bar and the avatars of its members, so overall status is visible at a glance without opening each board.

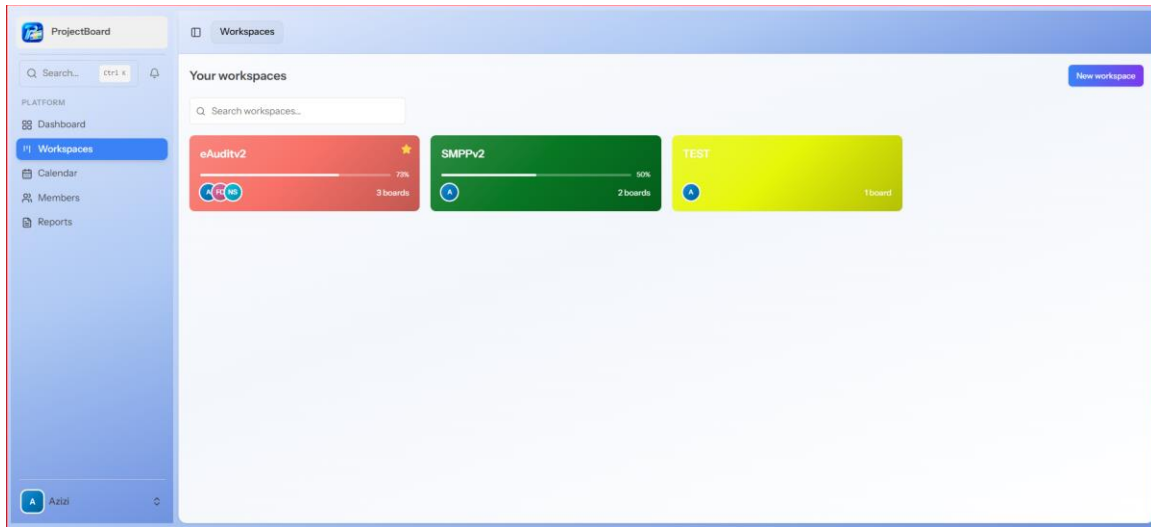


Figure 2: Workspace listing — progress bar and member avatars on each tile

- Search and pagination across workspaces and boards.
- Favourite a board or workspace to pin it to the top of the list.
- Archive boards, lists, and cards without permanently deleting them.

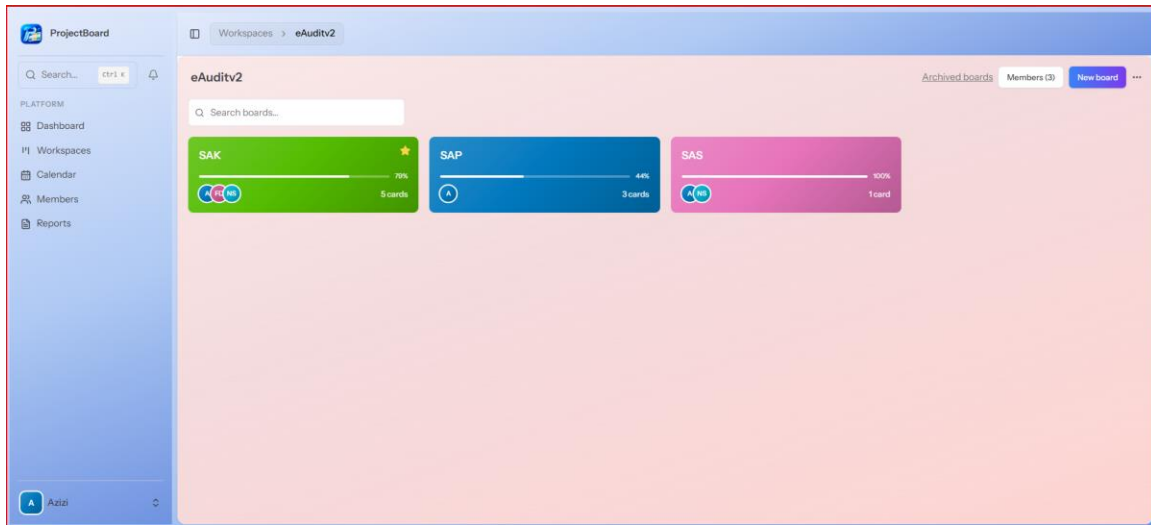


Figure 3: List of boards inside a workspace

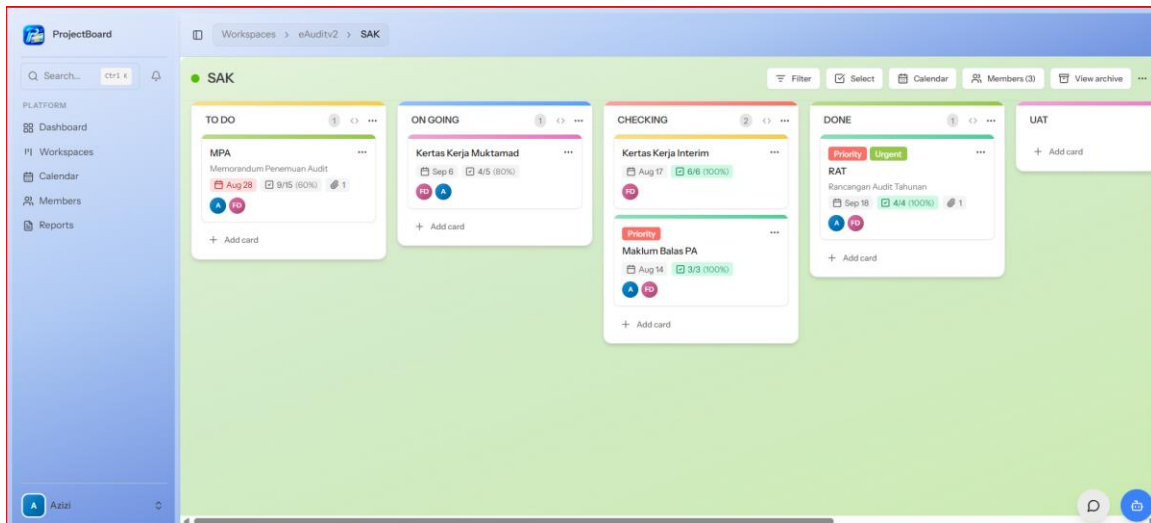


Figure 4: Kanban board view with lists and cards

3. Cards and Checklists

Each card can carry:

- A name, description, and colour.
- A due date, with overdue cards highlighted in red.
- One or more assigned members.
- Labels for categorisation.
- File attachments, with an optional cover image.
- One or more checklists, each with its own items.

Checklist items support their own due date, assigned member, and completion date (recorded automatically the moment an item is checked). This lets a team see not just whether a task was done, but when it was done relative to its deadline. A full activity log and comments (including @mentions) are also available on every card.

The screenshot displays the 'Edit card' modal for a card titled 'MPA' (Memorandum Penemuan Audit). The modal is divided into several sections:

- Card Information:** Title 'MPA' and description 'Memorandum Penemuan Audit'.
- Tools:** Buttons for 'Members', 'Labels', 'Dates', 'Checklist', 'Attachment', 'Color', and 'Cover'.
- Due Date:** A date picker set to 'Aug 28, 2026'.
- Assignees:** Two assignees are listed: 'A' (Azizi) and 'FD' (Frieda Block DDS).
- Checklist Items:**
 - GPPK100:** 80% complete. Items include 'UI' (checked), 'FORM' (checked), 'CREATE' (checked), 'UPDATE' (checked), and 'DELETE' (unchecked). Due dates are shown for each item: 'Aug 31' for UI, 'Aug 26' for FORM, 'Sep 4' for CREATE, and 'Aug 26' for UPDATE.
 - GPPK200:** 100% complete. Items include 'UI' (checked), 'FORM' (checked), 'CREATE' (checked), 'UPDATE' (checked), and 'DELETE' (checked). All items have a due date of 'Sep 3'.
 - GPPK300:** 0% complete. Items include 'UI' (unchecked), 'FORM' (unchecked), 'CREATE' (unchecked), 'UPDATE' (unchecked), and 'DELETE' (unchecked).
- Attachments:** One attachment is shown: 'projectboardText.png' (412.8 KB).
- Comments and activity:** A scrollable list of comments and activities. Recent comments include:
 - 'Azizi completed UPDATE in GPPK100 on this card' (Sep 4, 2026, 9:10 AM).
 - 'Azizi marked UPDATE incomplete in GPPK100 on this card' (Sep 4, 2026, 9:10 AM).
 - 'Frieda Block DDS completed DELETE in GPPK200 on this card' (Sep 3, 2026, 12:17 PM).
 - 'Frieda Block DDS completed UPDATE on this card' (Sep 3, 2026, 12:13 PM).
 - 'Azizi completed CREATE on this card' (Sep 3, 2026, 12:12 PM).
 - 'Azizi completed FORM on this card' (Sep 3, 2026, 12:12 PM).
 - 'Azizi set the due date to Aug 28, 2026' (Sep 3, 2026, 12:02 PM).
 - 'Azizi completed UI on this card' (Sep 3, 2026, 11:49 AM).
 - 'Azizi completed UPDATE on this card' (Sep 3, 2026, 11:49 AM).
 - 'Azizi marked UPDATE incomplete on this card' (Sep 3, 2026, 11:49 AM).
 - 'Azizi marked DELETE incomplete on this card' (Sep 3, 2026, 10:02 AM).
 - 'Azizi completed DELETE on this card' (Sep 3, 2026, 10:02 AM).
 - 'Azizi set the due date to Aug 27, 2026' (Sep 3, 2026, 10:00 AM).
 - 'Azizi @Frieda Block DDS ok bos' (Aug 26, 2026, 11:04 AM).
 - 'Frieda Block DDS @Azizi ok i already look' (Aug 26, 2026, 10:57 AM).
 - 'Azizi @Frieda Block DDS try look at this' (Aug 26, 2026, 10:54 AM).
 - 'Azizi added projectboardText.png to this card' (Aug 26, 2026, 10:12 AM).
 - 'Azizi marked DELETE incomplete on this card' (Aug 24, 2026, 3:57 PM).
 - 'Azizi completed DELETE on this card' (Aug 24, 2026, 3:57 PM).
 - 'Azizi completed UPDATE on this card' (Aug 24, 2026, 3:57 PM).
 - 'Azizi completed CREATE on this card' (Aug 24, 2026, 3:57 PM).
 - 'Azizi completed FORM on this card' (Aug 24, 2026, 3:57 PM).
 - 'Azizi completed UI on this card' (Aug 24, 2026, 3:57 PM).
 - 'Azizi moved this card from ON GOING to TO DO' (Aug 24, 2026, 3:28 PM).

Figure 5: Card detail modal — checklist, due dates, assignees, and activity log/comments

4. Roles and Access Control

Every board member has one of three roles:

- Viewer — read-only access; can still chat and view everything.
- Editor — full editing rights, except changing due dates.
- HOD (Head of Department) — everything an Editor can do, plus the exclusive ability to set or change due dates on cards and checklist items. Only the board owner can promote a member to HOD.

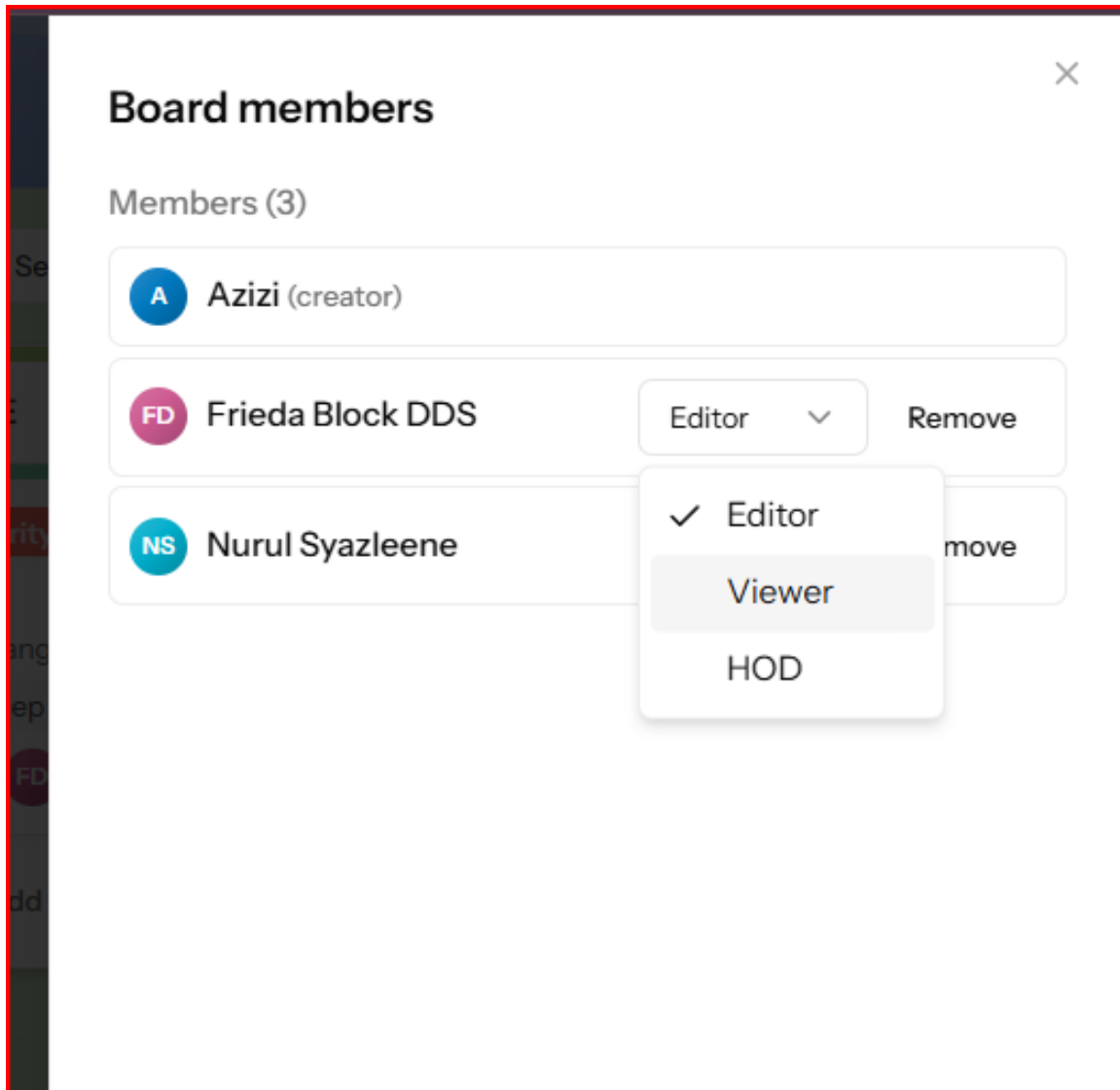


Figure 6: Board members panel with the Editor / Viewer / HOD role selector

5. Calendar

A calendar view aggregates every card and checklist-item due date across a board, or across all boards, into a single month view — useful for spotting deadline clusters ahead of time.

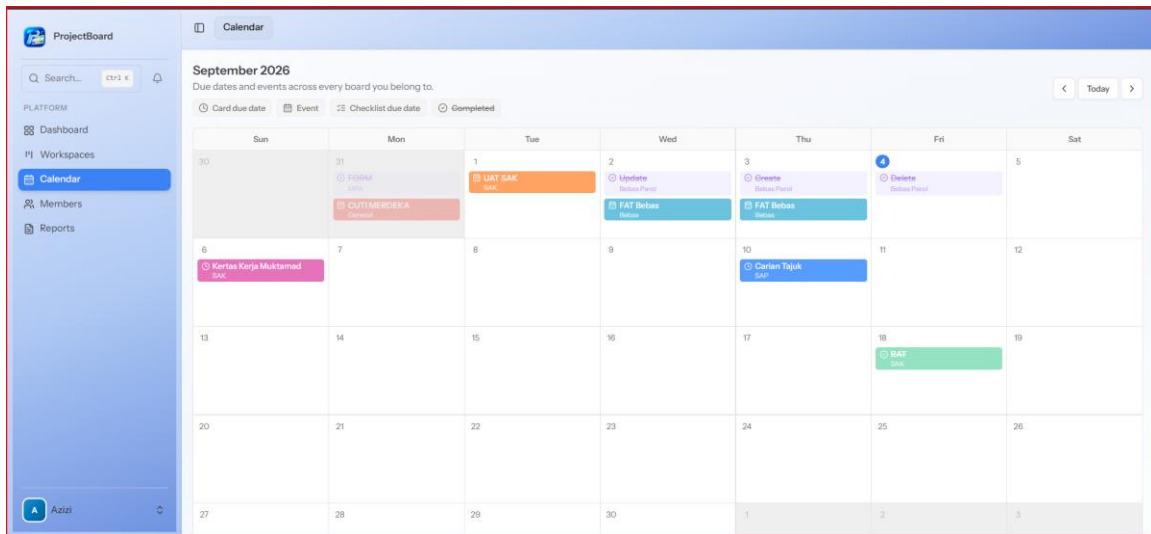


Figure 7: Calendar view showing due dates across boards

6. Dashboard

The Dashboard summarises a user's workload across every board they belong to: total tasks, completed, overdue, due-soon, checklist progress, tasks by board, and workload per member. Results can be filtered to a specific workspace or board, and downloaded as a PDF report.

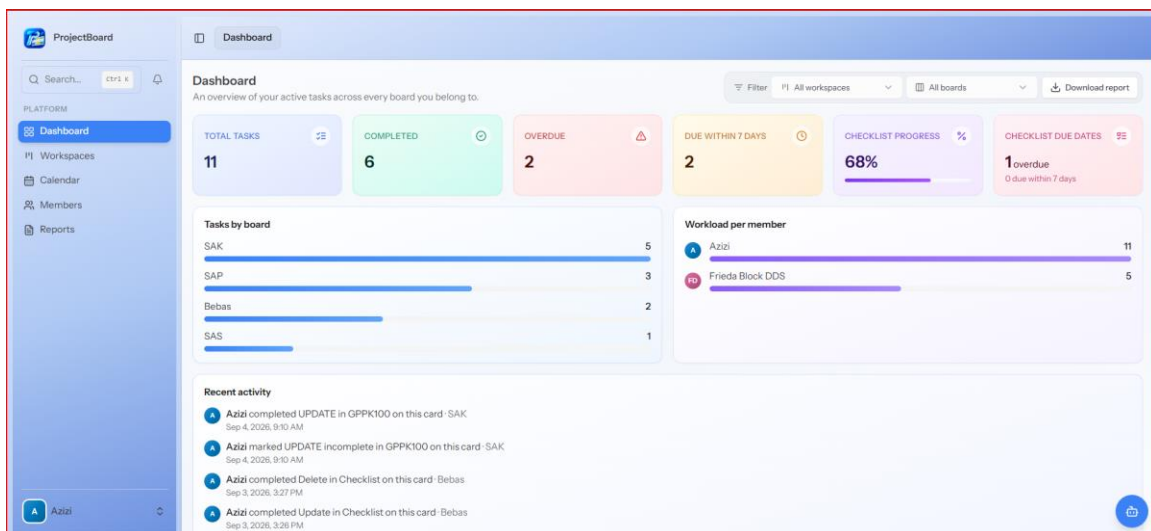


Figure 8: Dashboard page with statistics and charts

7. Reports

A dedicated Reports page offers five downloadable, print-ready PDF reports (one also exports to CSV), each scoped by workspace or board.

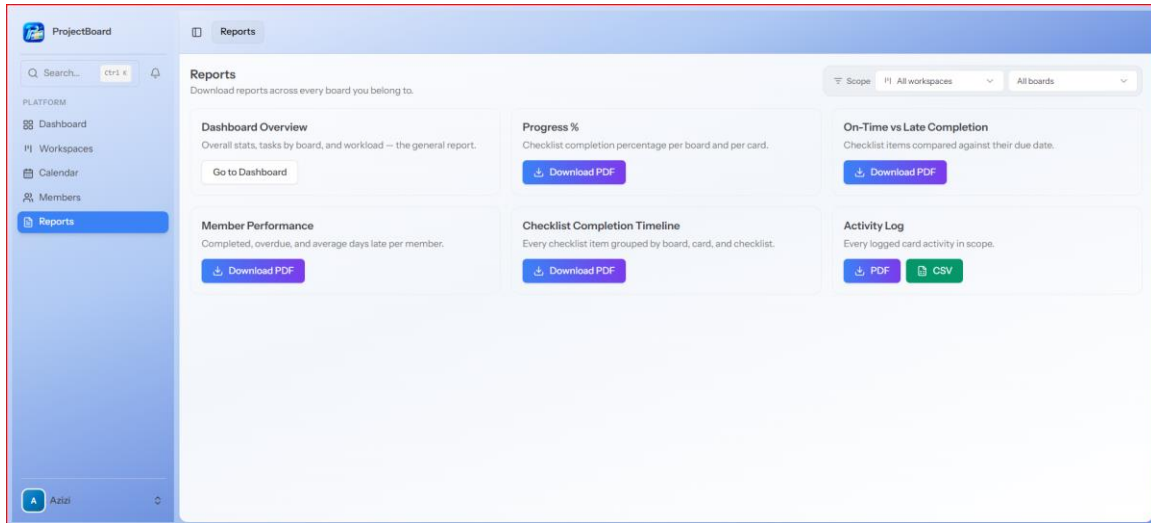


Figure 9: Reports page — five report types and a scope filter

7.1 Progress % — checklist-completion percentage per workspace, per board, and per card, with a colour-coded progress bar (red under 40%, blue 40–99%, green at 100%).

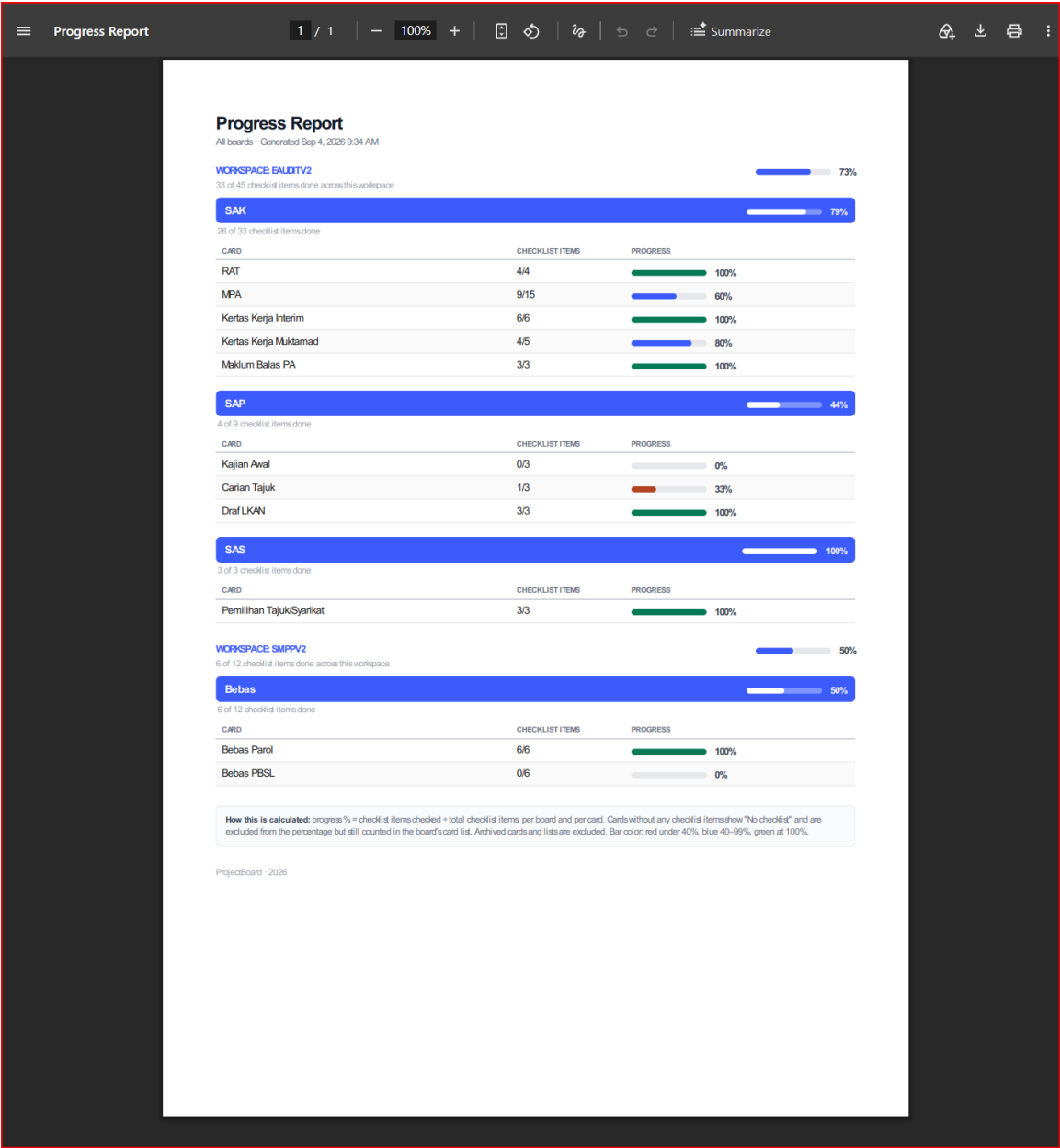


Figure 10: Sample Progress % report

7.2 On-Time vs Late Completion — compares each checklist item’s completion date against its due date, listing both on-time and late items with days-late and assignee.

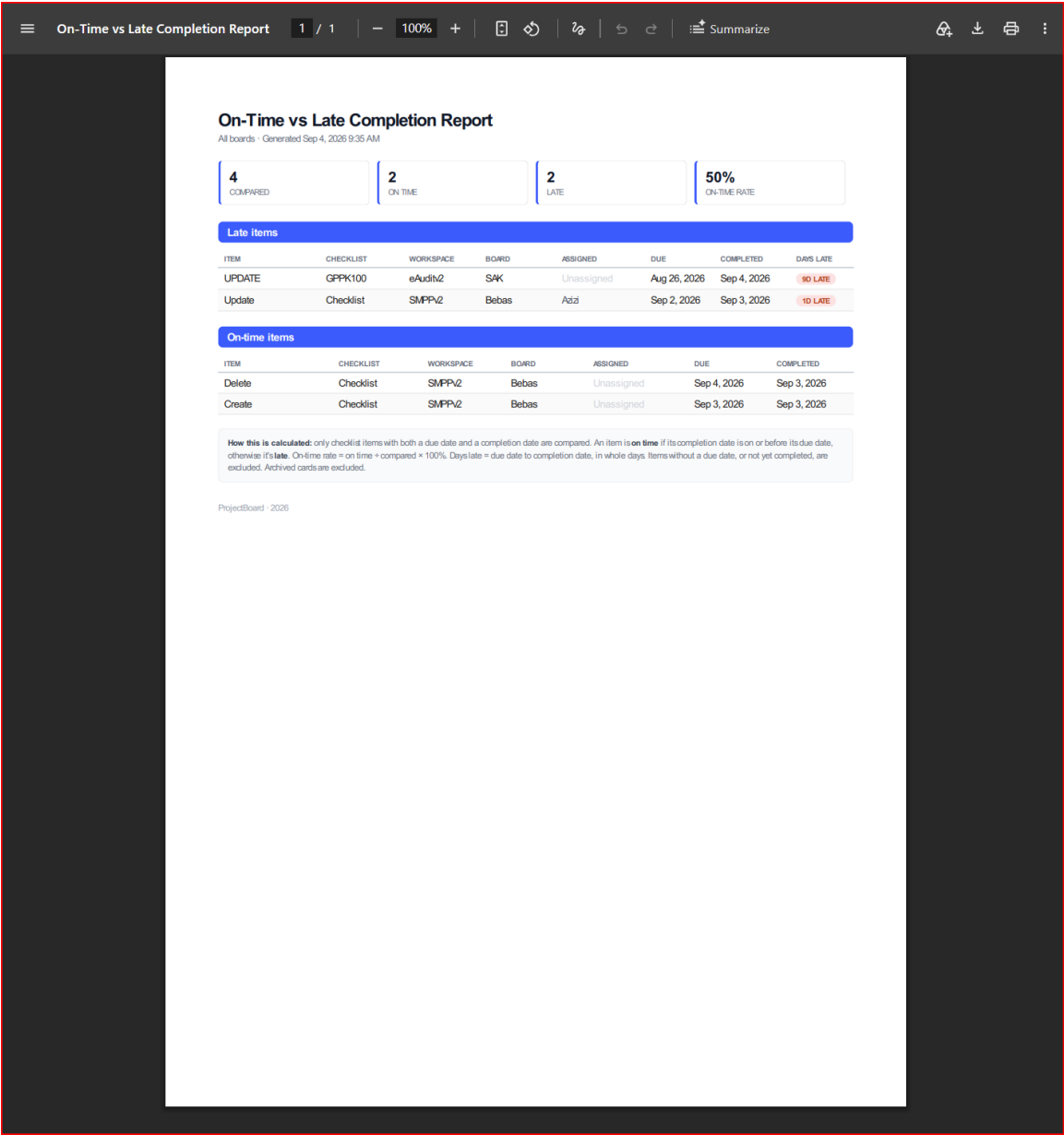


Figure 11: Sample On-Time vs Late Completion report

7.3 Member Performance — completed / pending / overdue counts and average days late per member, plus a full breakdown of every task assigned to each person.

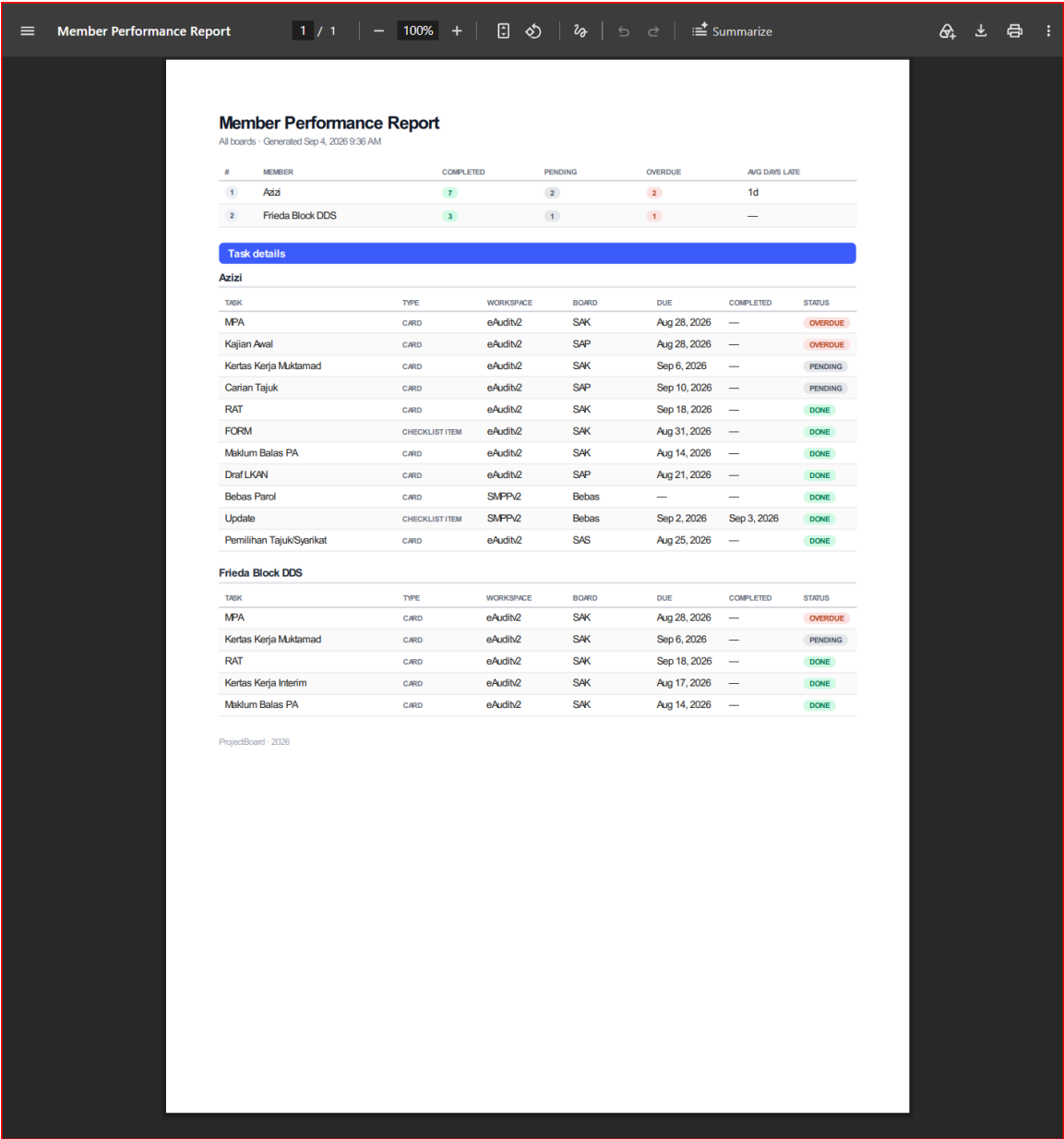


Figure 12: Sample Member Performance report

7.4 Checklist Completion Timeline — every checklist item grouped by workspace, board, card, and checklist, so items with the same name on different checklists are never confused.

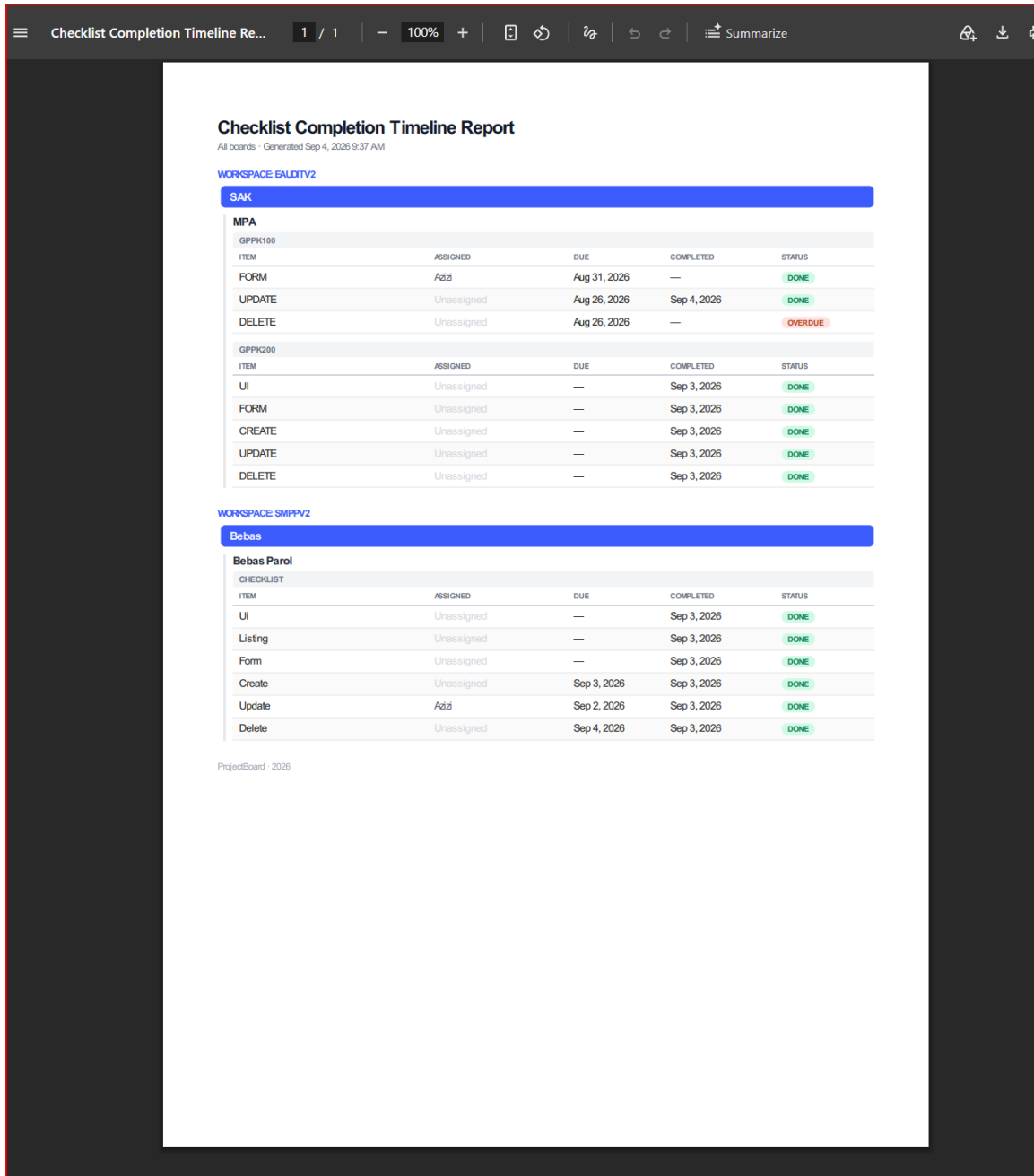


Figure 13: Sample Checklist Completion Timeline report

7.5 Activity Log — a chronological export of every logged action on a board (PDF and CSV).

Activity Log Report				
All boards - Generated Sep 4, 2026 9:37 AM				
WHEN	WORKSPACE	BOARD	USER	ACTIVITY
Sep 4, 2026 9:10 AM	eAudit2	SAK	Adzi	completed UPDATE on MPA
Sep 4, 2026 9:10 AM	eAudit2	SAK	Adzi	marked UPDATE incomplete on MPA
Sep 3, 2026 3:27 PM	SMFP2	Bebas	Adzi	completed Delete on Bebas Parol
Sep 3, 2026 3:26 PM	SMFP2	Bebas	Adzi	completed Update on Bebas Parol
Sep 3, 2026 3:16 PM	SMFP2	Bebas	Adzi	completed Create on Bebas Parol
Sep 3, 2026 3:15 PM	SMFP2	Bebas	Adzi	completed Form on Bebas Parol
Sep 3, 2026 3:15 PM	SMFP2	Bebas	Adzi	completed Listing on Bebas Parol
Sep 3, 2026 3:15 PM	SMFP2	Bebas	Adzi	completed UI on Bebas Parol
Sep 3, 2026 12:17 PM	eAudit2	SAK	Frieda Block DDS	completed DELETE on MPA
Sep 3, 2026 12:13 PM	eAudit2	SAK	Frieda Block DDS	completed UPDATE on MPA
Sep 3, 2026 12:12 PM	eAudit2	SAK	Adzi	completed CREATE on MPA
Sep 3, 2026 12:12 PM	eAudit2	SAK	Adzi	completed FORM on MPA
Sep 3, 2026 12:02 PM	eAudit2	SAK	Adzi	set the due date on MPA to Aug 28, 2026
Sep 3, 2026 11:49 AM	eAudit2	SAK	Adzi	completed UI on MPA
Sep 3, 2026 11:49 AM	eAudit2	SAK	Adzi	completed UPDATE on MPA
Sep 3, 2026 11:49 AM	eAudit2	SAK	Adzi	marked UPDATE incomplete on MPA
Sep 3, 2026 10:02 AM	eAudit2	SAK	Adzi	marked DELETE incomplete on MPA
Sep 3, 2026 10:02 AM	eAudit2	SAK	Adzi	completed DELETE on MPA
Sep 3, 2026 10:00 AM	eAudit2	SAK	Adzi	set the due date on MPA to Aug 27, 2026
Sep 2, 2026 5:01 PM	eAudit2	SAK	Adzi	set the due date on Kertas Kerja Muktamad to Sep 6, 2026
Aug 28, 2026 11:14 AM	eAudit2	SAK	Adzi	completed Delete on Kertas Kerja Interim
Aug 28, 2026 11:14 AM	eAudit2	SAK	Adzi	marked Delete incomplete on Kertas Kerja Interim
Aug 28, 2026 11:14 AM	eAudit2	SAK	Adzi	completed Delete on Kertas Kerja Interim
Aug 28, 2026 11:14 AM	eAudit2	SAK	Adzi	marked Delete incomplete on Kertas Kerja Interim
Aug 28, 2026 11:13 AM	eAudit2	SAK	Adzi	completed Delete on Kertas Kerja Interim
Aug 28, 2026 11:13 AM	eAudit2	SAK	Adzi	marked Delete incomplete on Kertas Kerja Interim
Aug 28, 2026 11:13 AM	eAudit2	SAK	Adzi	completed Delete on Kertas Kerja Interim
Aug 28, 2026 11:13 AM	eAudit2	SAK	Adzi	marked Delete incomplete on Kertas Kerja Interim
Aug 27, 2026 10:58 AM	eAudit2	SAK	Adzi	moved Kertas Kerja Muktamad from CHECKING to ON GOING
Aug 27, 2026 10:58 AM	eAudit2	SAK	Adzi	moved Kertas Kerja Muktamad from ON GOING to CHECKING
Aug 26, 2026 11:06 AM	eAudit2	SAK	Frieda Block DDS	added Adzi to Kertas Kerja Muktamad
Aug 26, 2026 11:04 AM	eAudit2	SAK	Adzi	commented on MPA
Aug 26, 2026 10:57 AM	eAudit2	SAK	Frieda Block DDS	commented on MPA
Aug 26, 2026 10:54 AM	eAudit2	SAK	Adzi	commented on MPA
Aug 26, 2026 10:12 AM	eAudit2	SAK	Adzi	added projectboardText.png to MPA
Aug 24, 2026 5:26 PM	eAudit2	SAS	Adzi	completed Lulus Pemilihan Tajuk/Syarikat on Pemilihan Tajuk/Syarikat
Aug 24, 2026 5:26 PM	eAudit2	SAS	Adzi	marked Lulus Pemilihan Tajuk/Syarikat incomplete on Pemilihan Tajuk/Syarikat
Aug 24, 2026 4:23 PM	eAudit2	SAS	Nurul Syazeene	moved Pemilihan Tajuk/Syarikat from CHECKING to COMPLETED

Figure 14: Sample Activity Log report

8. AI Assistants

ProjectBoard integrates Google Gemini in two places, both scoped to read the user's own data only.

8.1 Board Assistant — a chat widget on the board page that helps brainstorm board structure. It can propose new lists or cards, but a member with edit rights must explicitly approve them (by clicking “Add to board”) before anything is created — the AI never edits the board on its own.

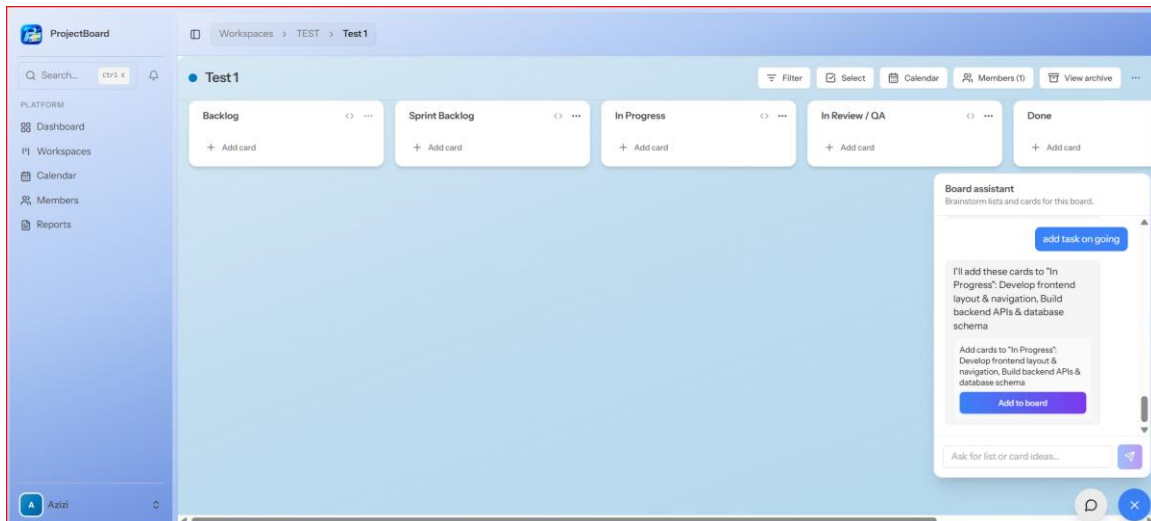


Figure 15: AI Board Assistant widget proposing new cards

8.2 Dashboard Assistant — a read-only chat widget that answers questions about overall progress using the same data shown on the Dashboard, and links directly to the relevant board when it mentions one. It replies in the same language the user writes in.

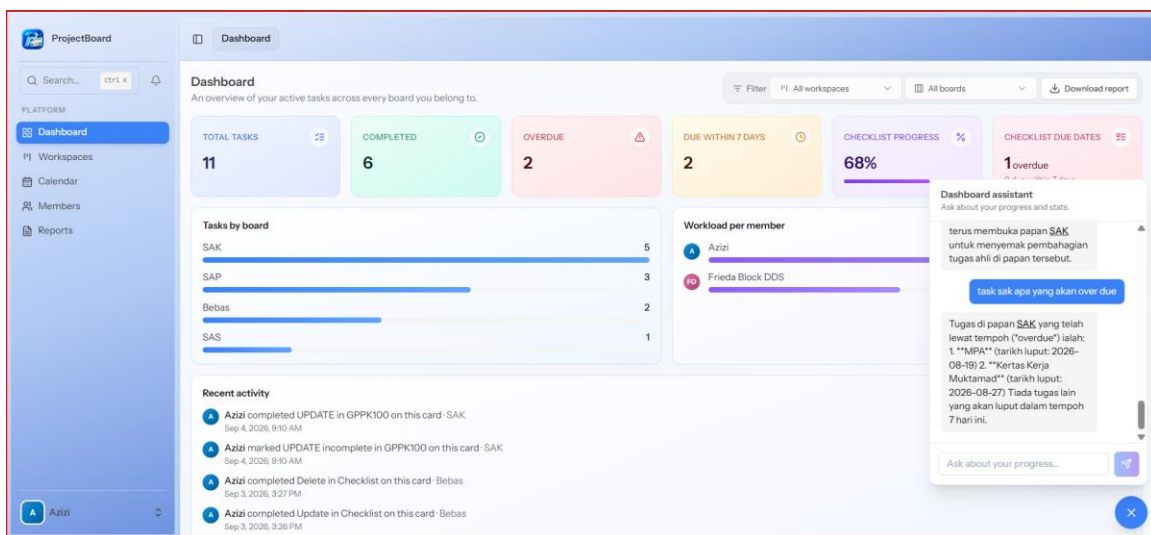


Figure 16: AI Dashboard Assistant widget replying in the user's own language

9. Board Chat

Every board has a shared group chat, open to all members regardless of role, with @mention support that notifies the mentioned member. Unread messages are shown as a badge on the chat icon.

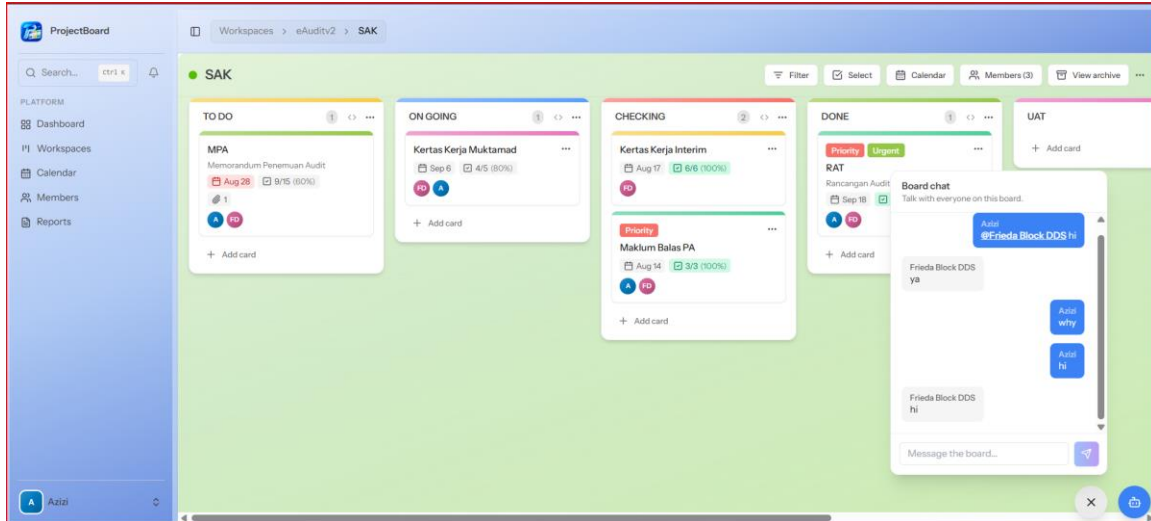


Figure 17: Board Chat widget with @mention

10. Notifications

Members are notified when they are assigned to a card or checklist item, or mentioned in a comment or board chat message. Notifications can be searched, filtered by read status, and marked as read individually or all at once.

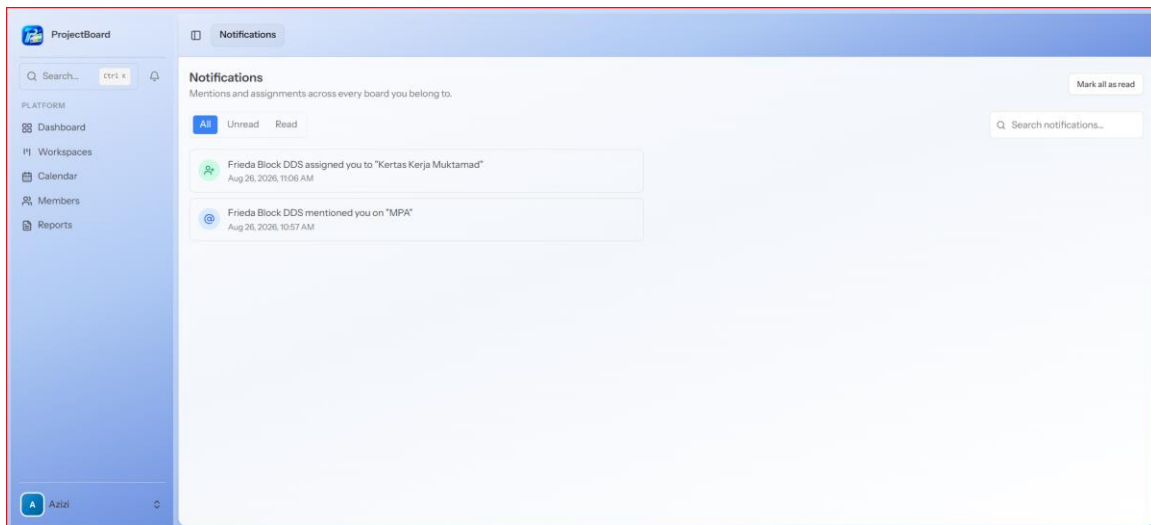


Figure 18: Notifications page

11. Advantages Over Paid Kanban Products

ProjectBoard is designed as a self-hosted alternative to commercial kanban systems such as Trello, Asana, and Jira. The comparison below covers cost, data ownership, and key features.

11.1 Cost

System	Cost	Notes
ProjectBoard	RM 0	Forever, no user-count limit
Trello Standard	~RM 24 / month	Per user
Trello Premium	~RM 47 / month	Per user
Asana / Jira	Comparable to Trello	Basic features gated behind paid tiers

The AI features (Board Assistant and Dashboard Assistant) run on Google Gemini's free tier — no additional cost, unlike Trello's Butler AI, which requires upgrading to a Premium/Business plan.

11.2 Data Ownership and Control

- Not third-party cloud — every board, card, and attachment is stored in your own database.
- Backups, migrations, or audits can be performed at any time, with no dependency on an external company's policies.
- The source code is your own — internal logic can be customised to match the organisation's real workflow.

11.3 On Par With, or Better Than, Trello Premium

Feature	Trello Premium	ProjectBoard
Checklist item	Due date only	Due date + assignee (PIC) + completion date
AI Assistant	Butler AI, limited credits	Free, no credit limit
Access roles	Viewer / Editor	Viewer / Editor / HOD (due-date control)
Analytics & Reports	Dashboard view only	Dashboard + 5 downloadable PDF/CSV reports
Calendar view	Yes	Yes, on par
Cost	~RM 47 / user / month	RM 0

11.4 No Scale Limits

Aspect	Trello Free	ProjectBoard
Number of boards	10 boards / workspace	No limit
Attachment size	10 MB / file	Limited only by server capacity
AI features	Requires Premium/Butler	Free, built in
Viewer role	Requires Business plan	Free, built in

12. Security and Data Handling

- All report exports (CSV) are sanitised against spreadsheet formula injection.
- File attachments are restricted to an allow-list of safe MIME types.
- Authentication routes (registration, password reset) are rate-limited.
- Cross-board data isolation is enforced and covered by automated tests — a user can never see another user's board data through any report or page.
- Password strength requirements are enforced at registration and reset, with a live strength meter.

13. Summary

ProjectBoard delivers the core kanban workflow teams expect — boards, cards, checklists, due dates, and collaboration — alongside features often reserved for paid tiers of commercial tools: due-date tracking down to the checklist item, role-based due-date control, AI-assisted planning and reporting, and a full suite of exportable reports. Being self-hosted, the organisation retains full control of its own data with no per-seat licensing cost.